

# Project: Authorship Verification

**Satoshi Nakamoto is the name used by the presumed pseudonymous person or persons who developed bitcoin.** Nakamoto's true identity is unknown. Various people have been posited as the person or group of people behind his name.



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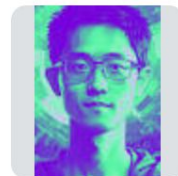
## Wei Dai | The Hunt For Satoshi

For this project:

Candidate author: Wei Dai

Data: 543 forum posts by Nakamoto, 230 forum posts and blogs by Dai

**Dai** is a Cypherpunk and his work on **B-Money** is the first to be referenced in the **Bitcoin** whitepaper. He is an expert C++ developer, which is the same ... [Read more](#)



\* Data provided in the course: LDA-T314 Language Technology for Forensic & Investigative Linguistics, University of Helsinki, 2025

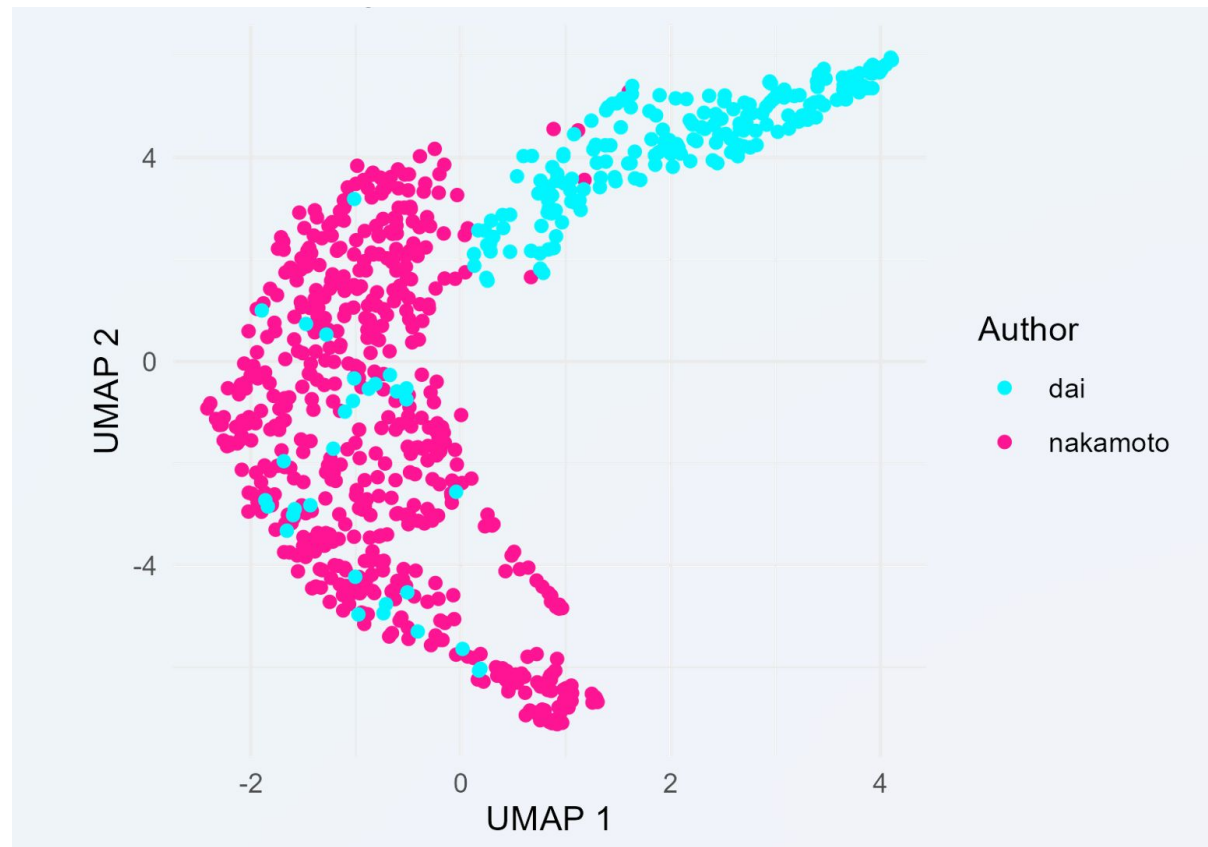
# 1. Jaccard Distance

A measure of dissimilarity between two sets, based on their shared and unique elements

In this case: measures the dissimilarity between texts based on their unique word types (which words are used), regardless of frequency

$$d_J(A, B) = 1 - \frac{|A \cap B|}{|A \cup B|}$$

If two texts have zero words in common, the distance is 1; if they have all words in common, the distance is 0

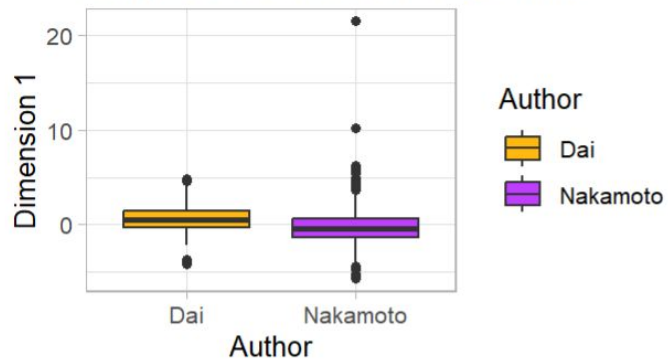


# Multidimensional Analysis (MDA)

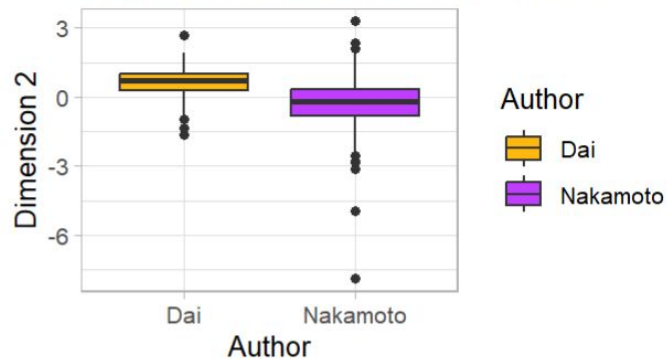
MDA takes many linguistic features, identifies patterns in how they vary together, and reduces them to a smaller number of dimensions that can be used to compare texts.

- Identify major parameters of linguistic variation based on co-occurrence
- Each dimension represents a set of linguistic features that co-occur

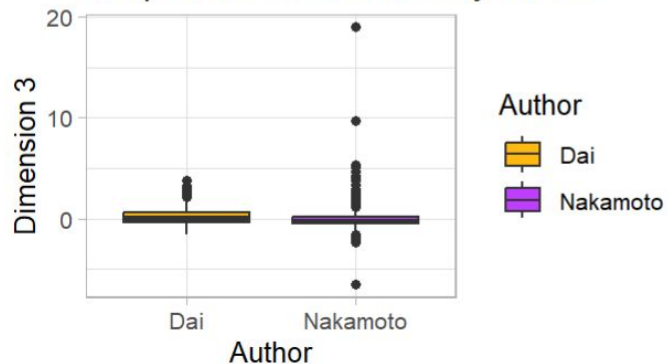
Boxplot of Dimension 1 by Author



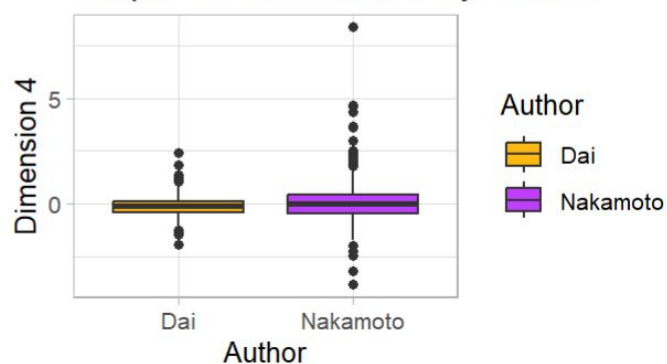
Boxplot of Dimension 2 by Author



Boxplot of Dimension 3 by Author



Boxplot of Dimension 4 by Author



Distribution of factor scores across four linguistic dimensions for Dai and Nakamoto's texts

# Possible methods with LLM

## 1. Classification-based

- Train one model on texts from multiple known authors
- Predicted probability for each candidate author

## 2. Perplexity-based

- Train separate language models on each candidate's texts
- Run the target text through both models and calculate perplexity (PPL)